Prediction and Its Uncertainty

Unit 4: Why Predictions Are Wrong, and How Wrong

Stat 220 | Statistics for Data Science

Where we left off

- Unit 3 built models and judged them on data they had not seen.
- Now we zoom all the way in on a single **prediction**: we hand the model one new case and ask for $\hat{y}$.
- The questions for this unit: when that prediction is wrong, *why* is it wrong, and *how wrong* should we have expected it to be?

Principle: a prediction is not a number

A prediction is a number *and* an honest range around it. A bare point estimate, with no sense of how far off it could be, is only half an answer.

Why this matters

- Every forecast, risk score, and "the model says \$420k" is a prediction someone will act on.
- The value isn't the point. It's knowing whether the real answer could be \$418k or \$250k, and *why*.
- Most prediction disasters are not bad point estimates. They are good point estimates with badly underestimated uncertainty.

In practice: the shape of this module

"Your model predicts this customer will spend \$500 next month. How wrong could that be, and where does the error come from?"

What this unit gives you

Things to know

- that a prediction is a point *plus* an honest interval
- the four layers of error: noise, estimation, model structure, and shift
- why a prediction interval is much wider than a confidence interval
- how to size an interval from real out-of-sample errors
- what extrapolation and regression to the mean do to predictions
- what it means for predictions to be **calibrated**

Mistakes to stop making

- reporting a point prediction with no interval at all
- giving intervals that are far too narrow
- quoting a confidence interval when the question needed a prediction interval
- extrapolating past the range of the data
- assuming next year looks like the training data
- mistaking regression to the mean for a real effect

Write a 90% interval for each

1. The length of the Nile River, in miles.
2. The number of bones in the adult human body.
3. The year the first iPhone went on sale.
4. The height of Mount Everest, in feet.
5. The number of countries in Africa.
6. The speed of light, in miles per second.

A low and a high for each. Wide enough to be 90% sure.

How wide a 90% interval has to be

1. Nile: about **4,130** miles
2. Bones: **206**
3. First iPhone: **2007**
4. Everest: **29,032** feet
5. Countries in Africa: **54**
6. Speed of light: $\approx$**186,000** mi/s

How many of your six intervals contained the truth?

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This is the whole module in one game: the point guess is easy, the honest range is hard.

How many of your six intervals contained the truth?

Trap: we are badly overconfident

Calibrated means about 5 or 6 hits out of 6. Most people get 2 or 3. Our intervals are almost always *too narrow*: we think we know more than we do.

Why a prediction is off: four layers

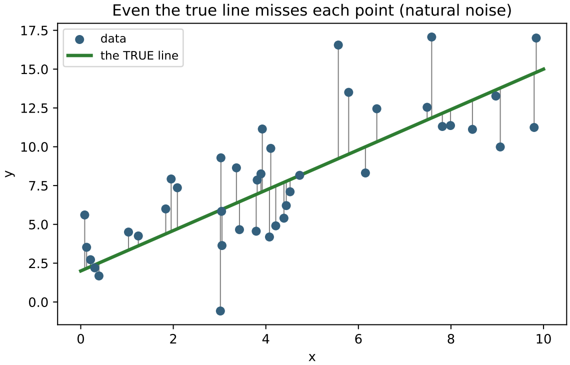
A prediction misses for reasons that stack on top of each other:

1. **Irreducible noise:** the world is random, so even a perfect model misses an individual.
2. **Estimation uncertainty:** we don't know the true model, we estimated it.
3. **Model error:** our model may be the wrong shape, or missing variables.
4. **Distribution shift:** the future may not look like the data we trained on.

Principle: know which layer you're fighting

Some layers shrink with more data, and some don't. Knowing which one dominates tells you whether to collect more data, change the model, or distrust the forecast entirely.

Layer 1: irreducible noise

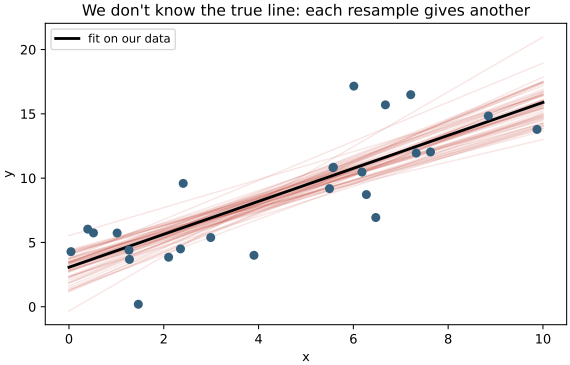


Even if you *knew* the true line exactly, individual points still scatter around it.

- This is the noise you can never remove.
- It sets a hard floor on how accurate any prediction of a single case can be.

More data does **not** shrink this layer.

Layer 2: estimation uncertainty



We never see the true line. We estimate it from a finite sample, and a different sample would give a slightly different line.

- Each faint red line is a fit from a resample of the data.
- Their spread is the uncertainty in the line itself.

This layer **does** shrink as n grows.

Layer 3: model (structural) error

- Maybe the truth is curved and we fit a line, or we left out a variable that matters.
- Then the model is **biased**: wrong in a systematic direction, not just noisy.
- Unlike estimation uncertainty, this does not go away with more data. A bigger sample of the wrong model just nails down the wrong answer more precisely.

Principle: the layer that hides

Noise and estimation error show up in the standard errors. Model error does not, because the software reports tight intervals around a biased prediction without complaint.

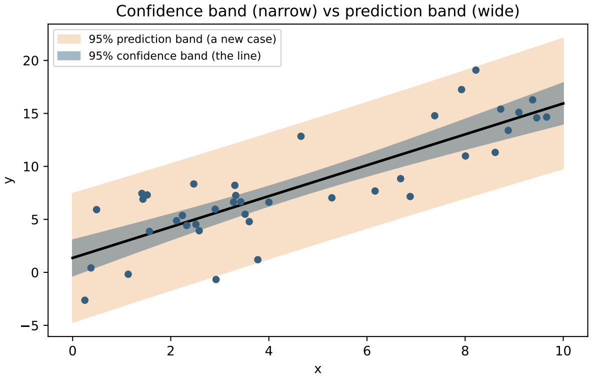
Layer 4: distribution shift

- Every model assumes the future is drawn from the same world as the training data.
- When that breaks (a new market, a new season, a pandemic, a changed competitor) the model can fail even though nothing about the math is wrong.
- This is the layer that wrecks real production systems most often, and no in-sample statistic can warn you about it.

In practice: a classic

"The model was 95% accurate in testing and tanked in production." Usually the answer is distribution shift or leakage, not a coding bug.

Putting it together: prediction band vs confidence band



- The **confidence band** covers the *line* (estimation only). Narrow.
- The **prediction band** covers a *new case*, so it adds the irreducible noise. Much wider.

Here the prediction band is about **six times wider** than the confidence band. Predicting a person is far harder than locating the average.

Your turn #1

Your turn: which layer dominates?

You're asked to predict **first-year sales of a brand-new product in a country you've never sold in**.

Turn to a neighbor: of the four layers (noise, estimation, model error, distribution shift), which one worries you most here, and why?

Your turn #1: answer

- **Distribution shift** and **model error** dominate: a new country and a new product are exactly the case your data does not cover.
- Estimation uncertainty (more data) and irreducible noise are real but secondary, and collecting more of the *old* data won't help.

Principle: the diagnosis changes the cure

If estimation is the problem, gather more data. If model error or shift is the problem, more of the same data is useless. You need a different model, new data, or a humble, very wide interval.

How to get honest error bars

- **Prediction intervals** bundle the noise and estimation layers into one range for a new case.
- **The bootstrap or an ensemble** (refit on many resamples) shows the estimation and model wobble directly, as the spread of predictions.
- **Held-out residuals** are the most honest check: how big were the errors on data the model never saw? That spread is your real uncertainty.

Principle: trust the errors you actually made

The most reliable error bar is empirical: look at how wrong the model was on fresh data, rather than at the interval the formula prints from the training set.

Building a prediction interval from your own errors

The most reliable interval needs no distributional theory at all:

1. Fit the model on training data.
2. Predict on a **held-out** set and collect the residuals $e_i = y_i - \hat{y}_i$.
3. For a new case, report $\hat{y} \pm$ the 5th and 95th *percentiles* of those held-out residuals.

- Worked: a model predicts 640 rentals. Held-out residuals run from -210 to $+260$ at the 5th and 95th percentiles. Report **640, with a 90% range of 430 to 900**.
- The interval is automatically the right width, because it is made of mistakes the model actually made.

Principle: why this beats the textbook formula

The formula assumes the model is correct and the noise is constant. Held-out residuals assume neither. They record what went wrong last time.

Your turn #2

Your turn: which interval does the question need?

Your model predicts revenue for a single store next month.

1. The CFO asks: "what will *this store* bring in?"
2. The board asks: "what is average monthly revenue *per store* across our 400 stores?"

With a neighbor: which question needs the wider interval, roughly how much wider, and why?

Your turn #2: answer

- (1) needs a **prediction interval**: it carries the store's own irreducible variability plus our uncertainty about the model.
- (2) needs a **confidence interval** for the mean, which shrinks like $1/\sqrt{400}$ and can be dramatically narrower.
- The prediction interval can easily be ten times wider. Handing the CFO the board's interval is the single most common way analysts get blamed for a "wrong" forecast.

Principle: one question to ask first

Am I predicting *one case* or locating *an average*? Everything about the width follows from that.

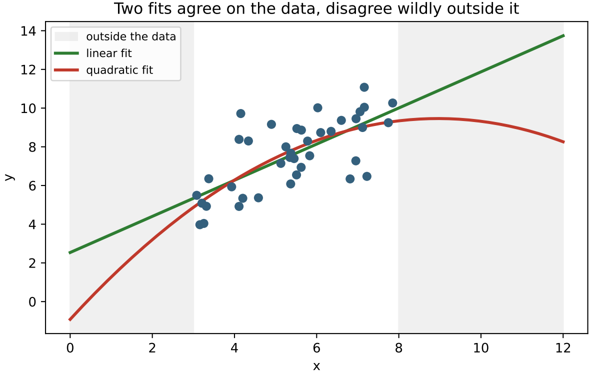
Calibration: do your intervals mean what they say?

- A 90% interval is only honest if, across many predictions, the truth lands inside about **90%** of the time.
- You can check this on held-out data: count the coverage. If your "90%" intervals catch the truth 60% of the time, they are too narrow, just like the calibration game.

Trap: confident and wrong

An overconfident model is worse than an honest one. Narrow intervals feel impressive in a meeting and cause disasters in production. Calibrated beats confident.

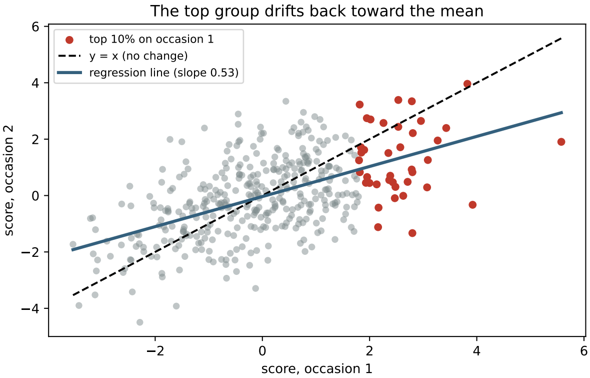
Folly 1: extrapolation



Two models can fit the data equally well and then **disagree wildly** once you leave the range you observed.

Mark Twain made the joke: extend the shrinking of the Mississippi River backward and "a million years ago" it was over a million miles long. "Wholesale returns of conjecture out of a trifling investment of fact."

Folly 2: regression to the mean



The top 10% on occasion 1 averaged **2.55**. Next time they averaged **1.29**, with *no* change in true ability.

- Extreme results are part skill, part luck, and luck doesn't repeat.
- This is the "sophomore slump" and the "cover jinx."

Folly 3: distribution shift, and a few more

- **Distribution shift:** a model trained on last year predicts this year badly, because the world moved.
- **Leakage:** a feature secretly contains the answer (or the future), so the model looks perfect in testing and useless in real life.
- **Goodhart's law:** once a predicted metric becomes a target, people game it and the prediction stops meaning anything.

Trap: too good to be true usually is

A model that predicts almost perfectly is more often a sign of leakage than of genius. Your first reaction to amazing accuracy should be suspicion, not celebration.

Scenario: the salesperson who "slumped"

Setup. Your top-performing rep last quarter had a much weaker second quarter. A manager says the bonus program ruined their motivation.

- Top performers are partly lucky, and luck doesn't repeat. A drop toward average is exactly what regression to the mean predicts, with no cause at all.
- The same logic flatters the bottom group: "the training fixed our worst reps," when they would have bounced back anyway.

You may be asked: how to answer

"Before blaming the bonus, I'd check whether this is just regression to the mean. To find a real effect I'd compare against a control group, not against the same people's lucky quarter."

LLM check: mistake or not? (1 of 2)

LLM check: we asked an LLM for a forecast

We told it: "Our model predicts $420k for this house. Give me a 95% interval."

It answered: "95% confident the price is $420k plus or minus $2k, so $418k to $422k."

Mistake, or no mistake?

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Mistake, or no mistake?

Trap: verdict: mistake

That \$2k is a confidence band on the *average* house at those features. A single house also carries irreducible noise, so the honest *prediction* interval is far wider, more like \$360k to \$480k. Quoting the CI as if it were the PI is dangerously overconfident.

LLM check: mistake or not? (2 of 2)

LLM check: same idea, a different answer

We told it: "We trained on 2019 data. Predict 2024 demand and tell me how much to trust it."

It answered: "The point forecast is X, but treat it cautiously: 2024 may differ from 2019 (distribution shift), so I'd widen the interval and validate on recent data before relying on it."

Mistake, or no mistake?

LLM check: mistake or not? (2 of 2)

LLM check: same idea, a different answer

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It answered: "The point forecast is X, but treat it cautiously: 2024 may differ from 2019 (distribution shift), so I'd widen the interval and validate on recent data before relying on it."

Mistake, or no mistake?

Principle: verdict: no mistake

This is the careful answer: it flags distribution shift, refuses to over-trust the point, and asks for recent validation. Exactly the instinct this module is building.

The arc of a prediction analysis

The order still matters more than the syntax:

1. **Name the target and the decision.** What are we predicting, and what will someone do with it?
2. **Build and check on held-out data**, never on the training fit.
3. **Attach an honest interval**, sized from real out-of-sample errors, not the training formula.
4. **Check calibration and shift:** do the intervals cover at the rate they claim, and is the future like the past?
5. **Communicate the uncertainty**, including extrapolation and "this could be very wrong if the world changes."

Producing a prediction with honest error bars

1. **Name the target and the decision.** What will someone do with this number?
2. **Fit on training data, predict on held-out data**, and keep those residuals.
3. **Take the 5th and 95th percentiles of the held-out residuals** and attach them to the point prediction.
4. **Check coverage** on data used for neither fitting nor calibration: do the 90% intervals catch 90%?
5. **Ask which layer dominates**: noise, estimation, model structure, or shift.
6. **Check whether you are extrapolating** past the range of the training data.
7. **Report the point, the interval, and one sentence on when it stops being valid.**

The traps, all in one place

Trap: the ones that bite most often

- Reporting a point prediction with no interval.
- Intervals that are too narrow (overconfidence).
- Quoting a confidence interval when you needed a prediction interval.
- Extrapolating past the range of the data.
- Assuming the future matches the training data.
- Mistaking regression to the mean for a real effect.
- Trusting amazing accuracy without checking for leakage.

Ten questions to be able to answer

1. A prediction is wrong. Walk me through the possible sources of the error.
2. What is the difference between irreducible noise and estimation uncertainty?
3. Which sources of error shrink with more data, and which do not?
4. Confidence interval or prediction interval: which do you give a customer, and why?
5. How would you actually put honest error bars on a forecast?
6. Your model was great in testing and failed in production. What happened?
7. Your best performer slumped the next quarter. Did your intervention fail?
8. A model predicts almost perfectly. Are you excited or suspicious?
9. Why is extrapolating beyond the data so dangerous?
10. What does it mean for intervals to be calibrated, and how would you check?

Mastery checklist

You have this unit when you can, from memory:

- report a prediction as a point plus an honest interval
- name the four layers of error and say which shrink with data
- explain why a prediction interval is wider than a confidence interval
- estimate uncertainty from held-out errors and the bootstrap
- recognize extrapolation, regression to the mean, leakage, and shift
- explain what calibration is and how to check it

How to report a prediction

Principle: the reusable structure

1. **The point:** the model's best guess.
2. **The interval:** honest, sized from real out-of-sample errors.
3. **The sources:** which layer dominates the uncertainty here?
4. **The boundaries:** are we extrapolating, and could the world have shifted?
5. **The caveat:** one sentence on when this prediction should not be trusted.

A point with no range is a guess. A point with an honest range is a forecast.

What you should be able to do with software

Nobody is asking you to memorize syntax. You are expected to know *which* procedure the question calls for, what to hand it, and what the output means. With an AI assistant open, you should be able to produce any of these and defend the result.

You should be able to	**What you would reach for**
Predictions from a fitted model	`model.predict(X_new)`
Held-out residuals	`y_test - model.predict(X_test)`
An empirical prediction interval	`np.percentile(resid, [5, 95])`
Confidence versus prediction band in OLS	`fit.get_prediction(X)` `.conf_int(obs=True)`
Coverage check	count how often the truth lands inside
Estimation wobble via resampling	refit on bootstrap samples and spread the predictions
Turn a predictive distribution into a decision	simulate outcomes, then average the cost

If you cannot say what the output means, the fact that you produced it is worth nothing.

Where we go next

- So far the prediction has been a number. **Unit 11** turns to categorical outcomes: rates, tables, and predicting a yes or no, where the model outputs a probability and somebody has to choose a threshold.
- The same ideas carry over: a probability is a prediction with uncertainty, it can be miscalibrated, and the *cost* of being wrong is rarely the same in both directions.

Principle: the through-line

Whether the output is a number or a yes/no, the job is the same: give your best guess, say how sure you are, and know when not to trust it.